



LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE

Final Term Examination 2025

Class: BSCS

Semester: III

Course Title: Calculus and Analytical Geometry

Course Code: GEDU-202

Max Marks: 50

Time allowed: 3hrs

Note: Attempt all questions

Attempt all long Questions:		
Q#1(a)	Find the formulas for $f \circ g$ and $g \circ f$, and state the domain of the composition, if $f(x) = \frac{1+x}{1-x}$, $g(x) = \frac{x}{1-x}$.	4
(b)	Find the formula for the inverse of $f(x) = \sqrt{3x-2}$ with x as an independent variable and state the domain of $f^{-1}(x)$.	4
(c)	Let $f(x) = \begin{cases} x-1, & x \leq 3 \\ 3x-7, & x > 3 \end{cases}$. Also find $\lim_{x \rightarrow 3} f(x)$.	3
Q#2(a)	Evaluate the following limits. (i) $\lim_{x \rightarrow +\infty} \frac{1-e^x}{1+e^x}$, (ii) $\lim_{h \rightarrow 0} \frac{\sin h}{1-\cos h}$, (iii) $\lim_{x \rightarrow -\infty} \frac{4x-1}{2x+5}$, (iv) $\lim_{x \rightarrow +\infty} \left(1 + \frac{3}{x}\right)^{-x}$.	10
Q#3(a)	Define continuity, what can you say about the continuity of the function $f(x) = \sqrt{9-x^2}$?	5
(b)	Find the constants k and m , if possible, that will make the function f continuous everywhere. $f(x) = \begin{cases} x^2 + 5, & x > 2 \\ m(x+1) + k, & -1 < x \leq 2 \\ 2x^3 + 7, & x \leq -1 \end{cases}$	5
Q#4(a)	Show that $y = x \sin x$ is a solution to the differential equation	3